

Deep Learning-based OSINT Threat Classification on Blockchain Infrastructure

Highlights

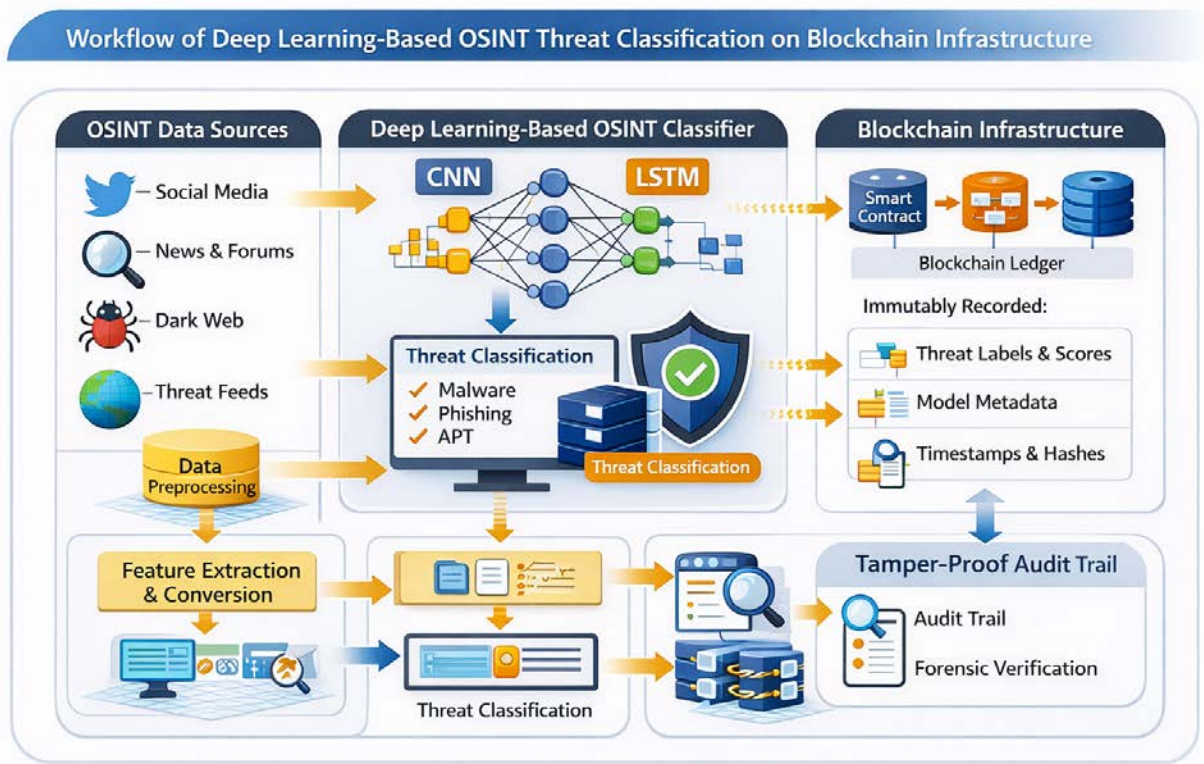
- The CNN–LSTM model accurately classifies heterogeneous cyber threat data.
- Blockchain integration ensures immutable logging and traceable decisions.
- The framework outperforms baseline models, achieving 96.4%, 95.8%, and 94.2% accuracy and 95.5%, 95.1%, and 93.4% F1-scores across the smart system, textual cyber threat, and Twitter datasets, respectively.

Abstract

The rapid growth of open-source intelligence (OSINT) has introduced new challenges in accurately classifying cyber threats while ensuring transparency and trust in automated decision-making. This paper proposes a novel deep learning-based OSINT threat classification framework deployed on a blockchain infrastructure to enable traceable and tamper-proof security analytics. Inspired by recent AI-driven OSINT models, the proposed approach integrates convolutional and long short-term memory (CNN–LSTM) networks to extract semantic and temporal features from heterogeneous OSINT data sources for effective threat identification. To address the lack of auditability in existing AI-based cyber defense systems, classification decisions, model metadata, and confidence scores are immutably recorded on a blockchain ledger using smart contracts. This design ensures integrity, non-repudiation, and post-incident forensic verification of AI outputs. The framework outperforms baseline models, achieving 96.4%, 95.8%, and 94.2% accuracy and 95.5%, 95.1%, and 93.4% F1-scores across the smart system, textual cyber threat, and Twitter datasets, respectively. Experimental evaluation demonstrates that the framework achieves reliable threat classification while significantly enhancing explainability and trust. The proposed system is well suited for security operations center (SOC) automation and accountable AI-driven cyber defense.

Keywords: OSINT, Deep learning, Blockchain security, Cyber threat classification

GRAPHICAL ABSTRACT



1. Introduction

The rapid digitalization of services and the widespread adoption of cloud-based platforms have significantly increased the complexity and frequency of cyber threats [1]. Modern cyber adversaries increasingly exploit open platforms such as social media, online forums, paste sites, and dark web marketplaces to coordinate attacks, share malware, and leak sensitive information [2] [3]. These openly accessible sources collectively referred to as Open-Source Intelligence (OSINT), have emerged as a critical asset for proactive cyber threat intelligence [4]. However, the sheer volume, velocity, and heterogeneity of OSINT data make manual analysis infeasible, necessitating automated and intelligent threat detection mechanisms [5].

Deep learning techniques have demonstrated strong potential in extracting high-level semantic patterns from unstructured OSINT data such as text streams, logs, and temporal threat feeds [6]. Models based on Convolutional Neural Networks (CNNs) effectively capture local spatial

features, while Long Short-Term Memory (LSTM) networks model sequential and temporal dependencies, making their hybrid integration suitable for OSINT threat classification tasks [7] [8]. Despite their effectiveness, most existing deep learning-based cyber intelligence systems operate as black boxes, offering limited transparency and traceability of decision-making processes [9] [10].

In security operations centers (SOCs), explainability, accountability, and forensic verifiability of AI-driven decisions are increasingly critical requirements [11]. Analysts must be able to trace how and when a threat classification decision was made, particularly in high-stakes environments such as financial systems, critical infrastructure, and national security networks [12] [13]. Conventional AI-based threat detection systems lack built-in mechanisms for immutable logging, making them vulnerable to tampering, repudiation, or post-incident disputes [14] [15].

Blockchain technology offers a promising solution to these challenges by providing decentralized, tamper-resistant, and verifiable data storage through immutable ledgers and smart contracts [16] [17]. Recent studies have explored blockchain for secure threat intelligence sharing and data integrity; however, its integration with real-time deep learning-based OSINT classification remains limited [18] [19]. In particular, the execution trace of AI models, including classification outcomes, confidence scores, and model metadata, is rarely recorded on-chain in existing systems [20].

To address these limitations, this paper proposes a deep learning-based OSINT threat classification framework integrated with blockchain infrastructure to enable traceable and auditable AI-driven decisions. By coupling a CNN–LSTM classifier with blockchain-based logging mechanisms, the proposed approach ensures secure storage of threat labels, timestamps, and model execution hashes. This integration enhances trust, transparency, and forensic readiness while maintaining high classification performance, thereby supporting advanced SOC automation and explainable cyber defense systems.

1.1 Contributions

The novel contributions of this study are:

1. Proposes a blockchain-anchored OSINT threat classification framework enabling tamper-proof tracking of deep learning decisions.
2. Integrates a CNN–LSTM model with smart-contract logging to achieve explainable and auditable cyber threat intelligence.
3. Demonstrates secure SOC-oriented automation by linking AI threat predictions with immutable blockchain audit trails.

2. Literature Review

This section reviews recent studies on OSINT-based threat intelligence, deep learning approaches for cyber threat classification, and the emerging role of blockchain in securing and auditing AI-driven cybersecurity systems. Table 1 shows summary of research gaps

Tolah et al. (2025) [21] propose BlockIntelChain, a blockchain-based cyber threat intelligence (CTI) sharing framework for IoT networks. It integrates a hybrid consensus mechanism with privacy-preserving techniques such as Differential Privacy, Zero-Knowledge Proofs, Homomorphic Encryption, and Secure Multi-Party Computation, alongside Federated Learning on edge nodes. Evaluations on MISP datasets show 923 TPS at 500 nodes, 99.6% consensus success, and FL-based models achieving 96.4% accuracy for malware, 94.7% for phishing, and 95.2% for network anomalies, outperforming centralized baselines.

Gwasssi et al. (2025) [22] develop Cyber-XAI-Block, an end-to-end cyber threat detection and risk assessment framework for IoT-enabled smart organizations. The system uses Capsule Network, Quantum Key Secure Communication, DQN-A3C reinforcement learning, and XAI models including SILT and SHAP. Experimental results show a 97% detection rate, 98% prevention accuracy, 45% false positive rate, and 97 s end-to-end response time, demonstrating superior performance across all metrics compared to existing methods.

Dale et al. (2025) [23] propose an AI-based framework for Twitter-based OSINT for cyber event analytics, automatically filtering tweets for cybersecurity information, aggregating events, and assessing exploitability. Evaluation demonstrates effective detection of cybersecurity events and improved situational awareness for operators.

Coutinho and de Araújo (2025) [24] introduce MICRA, a modular intelligent cybersecurity response architecture combining supervised learning, heuristic analysis, and behavioral modeling across six interoperable modules. The architecture supports dynamic data acquisition, threat analysis, multi-layer validation, and adaptive response orchestration, offering a scalable foundation for proactive cyber defense in diverse operational environments.

Presekhal et al. (2025) [25] present a hybrid deep learning-based anomaly detection system for cyber-physical power systems, integrating graph convolutional networks with LSTM layers and attack graphs. Evaluations demonstrate efficient identification and localization of anomalies in OT networks, providing robust detection of active cyber-attacks in near real time.

Khayat et al. (2025) [26] propose a blockchain-powered threat intelligence system combining graph convolutional autoencoders, reinforcement learning feedback, and secure blockchain logging. The system achieves high accuracy, precision, negative predictive value, and MCC, with low false positive and negative rates, validating reliable and scalable cyber threat detection.

Sufi (2025) [27] develops a GPT-based cyber threat assessment framework for industrial cyber-physical systems, integrating CNN, spectral residual transformation, and regression analysis. Evaluated on 9,018 cyber events, the system achieves precision 0.999, recall 0.998, F1-score 0.998, successfully detecting anomalies and providing actionable intelligence through knowledge graph analysis.

Naveeda and Fathima (2025) [28] implement IoT-E-CADS, a bi-level IoT-enabled cyberattack detection system for advanced metering infrastructure. Using Isolation Forest and Decision Tree ML algorithms, the system successfully detected two manually generated cyberattacks, achieving 95% accuracy in real-time monitoring and providing secure energy infrastructure surveillance.

Saeed et al. (2025) [29] propose an XAI-based reinforcement learning framework for mobile edge networks, leveraging A2C, PPO, and HER algorithms with SHAP explanations. The models achieve 99.97%, 99.98%, and 99.95% accuracy, reduce false positives by 30%, and outperform traditional ML methods, ensuring interpretable and adaptive threat detection.

Vajrobol et al. (2025) [30] present a machine learning approach for detecting IoT spoofing attacks, using Random Forest on 46 features. The system achieves 95.1% accuracy, 95.2% precision, and 95.1% F1-score, successfully detecting both DNS and ARP spoofing, highlighting robust protection for IoT environments.

Table 1: Summary of Research Gaps in Cyber Threat Intelligence Studies

Study	Focus / Approach	Key Results	Identified Research Gaps
Tolah et al. (2025) [21]	Blockchain-based CTI sharing for IoT with FL	923 TPS at 500 nodes; 99.6% consensus; 96.4% malware accuracy, 94.7% phishing, 95.2% network anomalies	Limited evaluation on heterogeneous OSINT data; scalability under dynamic IoT environments
Gwasssi et al. (2025) [22]	XAI + FL for smart organization threat detection	97% detection rate; 98% prevention accuracy; 45% FPR; 97 s response time	High false positive rate; reliance on specialized hardware; real-time deployment challenges
Dale et al. (2025) [23]	Twitter-based OSINT analytics	Effective aggregation and filtering of cybersecurity events	Lack of integration with structured threat intelligence; limited automated mitigation strategies
Coutinho & de Araújo (2025) [24]	Modular cybersecurity response architecture	Conceptual framework with six interoperable modules	Lacks empirical evaluation on large-scale operational networks
Presekal et al. (2025) [25]	Hybrid deep learning for power systems	Efficient anomaly detection in OT networks; near real-time localization of attacks	Focused on OT networks only; generalization to other CPS domains not addressed
Khayat et al. (2025) [26]	Blockchain + GCN autoencoder + RL	High accuracy, precision, MCC; low FPR & FNR	Complexity of model training; latency impact on real-time threat detection not quantified

Study	Focus / Approach	Key Results	Identified Research Gaps
Sufi (2025) [27]	GPT-based cyber threat assessment	Precision 0.999; Recall 0.998; F1-score 0.998	Limited real-time evaluation; dependency on large curated datasets
Naveeda & Fathima (2025) [28]	IoT-enabled cyberattack detection	95% accuracy; successful detection of manually generated attacks	Small-scale deployment; scalability and multi-attack scenarios underexplored
Saeed et al. (2025) [29]	XAI-RL for mobile edge networks	Accuracy: 99.97–99.95%; 30% reduction in false positives	High resource consumption; false negatives (42%); integration with heterogeneous IoT platforms needed
Vajrobol et al. (2025) [30]	ML for IoT spoofing detection	Accuracy 95.1%; Precision 95.2%; F1-score 95.1%	Limited attack types; performance on dynamic IoT networks and multi-modal attacks not evaluated

2.1 Research gaps

Recent studies on cyber threat intelligence and detection have highlighted several research gaps. Most frameworks focus on specific environments such as IoT networks, smart organizations, or cyber-physical systems, limiting their generalizability across heterogeneous OSINT and behavioral datasets. Several approaches demonstrate high detection accuracy but suffer from high false positive rates, reliance on specialized hardware, or small-scale deployments, restricting real-time applicability. Conceptual and modular architectures often lack empirical validation on large-scale or dynamic operational networks. Additionally, integration of structured and unstructured threat intelligence, scalability for multi-attack scenarios, and deployment on resource-constrained edge devices remain underexplored. Latency, computational overhead, and the effectiveness of privacy-preserving mechanisms in maintaining utility while ensuring security also present open challenges. Collectively, these gaps indicate the need for frameworks

that combine high accuracy, explainability, traceability, and scalability across diverse cyber threat contexts.

2.2 Problem Statement

The increasing reliance on OSINT-driven deep learning models for cyber threat detection has improved detection accuracy but introduced challenges related to transparency, trust, and post-incident accountability. Existing AI-based OSINT systems function as black boxes, offering limited traceability of classification decisions, which hinders forensic validation and regulatory compliance in Security Operations Centers (SOCs). Although blockchain technology provides immutability and auditability, its adoption for tracking AI model execution and threat classification decisions remains underexplored. Therefore, there is a critical need for an integrated framework that combines deep learning-based OSINT threat classification with blockchain-enabled audit trails to ensure secure, explainable, and tamper-proof cyber defense mechanisms.

3. Objectives

The novel objectives of this study are:

1. To design a deep learning-based OSINT threat classification model for accurate cyber threat detection.
2. To implement blockchain-based logging of AI inference decisions to ensure integrity and traceability.
3. To evaluate the effectiveness of explainable, tamper-proof AI decisions in supporting automated security operations.

4. Proposed Methodology

Fig 1 illustrates the end-to-end workflow of the proposed deep learning-based OSINT threat classification framework integrated with blockchain infrastructure. The process begins with the acquisition of heterogeneous OSINT data from social media, forums, blogs, dark web sources, and threat feeds. The collected data undergoes systematic preprocessing, including noise removal, tokenization, normalization, and encoding, to ensure consistency and model readiness.

Workflow of Deep Learning-Based OSINT Threat Classification on Blockchain Infrastructure

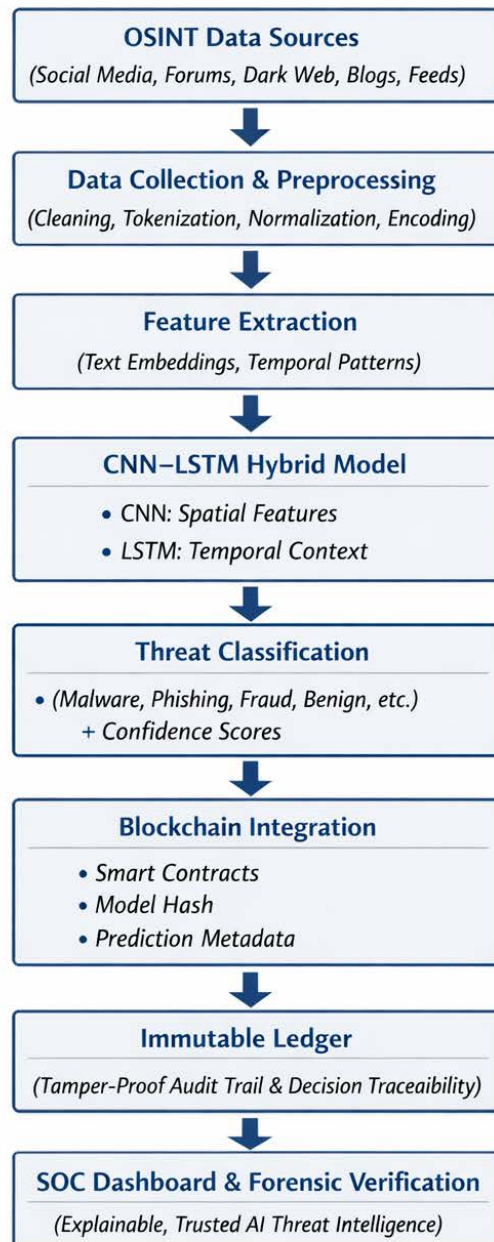


Fig 1: Workflow of Deep Learning-Based OSINT Threat Classification on Blockchain Infrastructure

Next, semantic and temporal features are extracted using text embeddings and time-dependent patterns. These features are processed by a hybrid CNN–LSTM model, where the CNN captures spatial and contextual representations, while the LSTM models sequential and temporal dependencies. The trained model classifies threats into multiple categories such as malware, phishing, fraud, or benign activities, along with associated confidence scores. To ensure transparency and trust, classification outputs, model hashes, and prediction metadata are recorded on the blockchain via smart contracts. The immutable ledger thus enables tamper-proof audit trails and decision traceability, which are finally visualized through a SOC dashboard to support explainable threat intelligence and forensic verification.

4.1 OSINT Data Acquisition

OSINT data acquisition forms the foundational stage of the proposed threat classification framework. Data are collected from diverse open-source intelligence channels, including social media platforms, online news portals, technical forums, dark web repositories, and structured threat intelligence feeds. These sources provide heterogeneous textual streams that capture emerging cyber threats, malicious activities, and contextual indicators relevant to security operations.

To ensure relevance and timeliness, data collection is performed continuously using automated crawlers and API-based extraction mechanisms. Each acquired OSINT instance is timestamped and associated with its source identifier to preserve provenance and temporal ordering. The collected raw data are denoted as a sequence of OSINT records:

$$D = \{d_1, d_2, \dots, d_n\}$$

where d_i represents the i^{th} OSINT data instance obtained from open sources. This dataset serves as the input for subsequent preprocessing, feature extraction, and deep learning-based threat classification stages.

4.2 Data Preprocessing and Normalization

Due to the noisy and unstructured nature of OSINT data, a comprehensive preprocessing pipeline is applied prior to feature extraction. This stage includes noise removal, duplicate elimination, tokenization, stop-word removal, lemmatization, and normalization to standard textual forms. URLs, emojis, special characters, and irrelevant symbols are filtered to reduce dimensionality and enhance semantic consistency. The cleaned text is then transformed into a normalized representation suitable for deep learning models.

Formally, the preprocessing operation can be expressed as:

$$x_i^* = P(x_i)$$

where x_i is the raw OSINT text and x_i^* denotes the preprocessed output after applying the preprocessing function $P(\cdot)$. Normalization ensures uniform input length and vocabulary alignment, facilitating stable convergence of the CNN-LSTM model and improving classification robustness across diverse OSINT sources.

4.3 Feature Extraction and Representation

For a given preprocessed OSINT text sequence:

$$T = \{\omega_1, \omega_2, \dots, \omega_n\}$$

each token ω_i is mapped to a dense vector representation using an embedding function $f(\cdot)$

$$e_i = f(\omega_i), e_i \in \mathbb{R}^d$$

The resulting embedded sequence is represented as:

$$E = [e_1, e_2, \dots, e_n]$$

To construct a fixed-length feature map suitable for deep learning, convolutional feature extraction is applied:

$$h_j = \sigma(W_c * E_{j:j+k-1} + b_c)$$

where W_c denotes the convolution kernel, k is the filter size, b_c is the bias term, and $\sigma(\cdot)$ is a nonlinear activation function.

For temporal OSINT streams, sequential feature representations are preserved and forwarded to recurrent layers to model time-dependent threat patterns. The final feature representation F serves as input to the CNN–LSTM hybrid threat classification model.

4.4 CNN–LSTM Hybrid Threat Classification Model

The CNN–LSTM hybrid model is designed to capture both **local semantic patterns** and **temporal dependencies** present in heterogeneous OSINT data. Convolutional Neural Network (CNN) layers are first employed to automatically learn discriminative spatial features from the extracted OSINT feature vectors, such as indicative keywords, phrases, or structural patterns related to cyber threats. These convolutional outputs are then passed to Long Short-Term Memory (LSTM) layers, which model sequential and temporal relationships, enabling the system to identify evolving threat behaviors over time. Finally, a fully connected layer followed by a Softmax activation function produces probabilistic threat class predictions. This hybrid architecture improves classification robustness for dynamic and noisy OSINT environments. Algorithm 1 shows proposed CNN–LSTM-Based OSINT Threat Classification Algorithm that forms the core deep learning inference pipeline used for OSINT threat classification prior to blockchain-based decision logging.

Algorithm 1: Proposed CNN–LSTM-Based OSINT Threat Classification Algorithm

Input: Preprocessed OSINT feature vectors $X = \{x_1, x_2, \dots, x_n\}$

Output: Threat class label y and confidence score

1. **Initialize model parameters**

Initialize convolutional filters W_c , LSTM weights W_l , and fully connected weights W_f .

2. **CNN-based spatial feature extraction**

Apply convolution and pooling to input feature matrix X :

$$F_c = Pool(\sigma(W_c * X + b_c))$$

where $*$ denotes convolution and $\sigma(\cdot)$ is the activation function.

3. **Temporal dependency learning using LSTM**

Feed CNN features to LSTM to model sequential patterns:

$$h_t = LSTM(F_c, h_{t-1})$$

4. **Hidden state representation generation**

Extract the final LSTM hidden state h_T as the threat representation.

5. **Threat probability estimation**

Apply a fully connected layer with Softmax activation:

$$P = Soft\ max(W_f h_T + b_f)$$

6. **Threat label assignment**

Select the threat class with maximum probability:

$$\hat{y} = \arg\max(P)$$

7. **Output prediction**

Return predicted threat label $\hat{y}$ and confidence score $\max(P)$.

4.5 Threat Labeling and Confidence Scoring

After feature learning through the CNN–LSTM hybrid model, the final classification layer produces a probability distribution over predefined threat classes using the Softmax function. Let the output logits from the fully connected layer be denoted as $z = [z_1, z_2, \dots, z_C]$, where C represents the number of threat categories. The Softmax function converts these logits into class probabilities as:

$$P(y = i | x) = \frac{e^{z_i}}{\sum_{j=1}^C e^{z_j}}$$

The predicted threat label $\hat{y}$ is assigned based on the maximum probability:

$$\hat{y} = \arg \max_i P(y = i | x)$$

The corresponding confidence score is defined as the highest Softmax probability associated with the predicted class. This confidence value quantifies the model's certainty in its decision and is later recorded along with the predicted label on the blockchain. Such confidence-aware labeling supports explainability and enables SOC analysts to prioritize alerts based on both threat severity and prediction reliability.

4.6 Blockchain-Based Decision Logging

Algorithm 2 ensures tamper-proof traceability of OSINT threat classification decisions by cryptographically hashing model outputs and recording them on a blockchain ledger.

Algorithm 2: Proposed Blockchain-Enabled Model Execution Tracking Algorithm

Input: Predicted threat label $\hat{y}$, confidence score C , timestamp t

Output: Immutable blockchain record for audit and forensic verification

1. Receive the predicted threat label $\hat{y}$ and confidence score C from the CNN-LSTM classifier.
2. Generate a cryptographic decision hash:

$$H = \text{Hash}(\hat{y} \parallel C \parallel t)$$

3. Create a blockchain transaction embedding the hash H along with minimal metadata.
4. Submit the transaction to the blockchain network for validation.
5. Append the validated transaction to the immutable distributed ledger.
6. Enable future audit and forensic verification by matching stored hashes with recomputed values.

4.7 Immutable Audit Trail and Traceability Mechanism

The immutable audit trail and traceability mechanism ensures transparency and trust in the OSINT threat classification process by permanently recording model decisions on a blockchain ledger. Each classification outcome, along with its associated confidence score and timestamp, is cryptographically hashed and stored as a blockchain transaction. Due to the inherent immutability of blockchain, once recorded, these decision logs cannot be altered or deleted, preventing post-incident tampering. During audits or forensic investigations, the stored hashes can be recomputed and compared with the original records to verify the authenticity and integrity of AI-driven decisions. This mechanism enables end-to-end traceability of model executions, supports accountability in SOC operations, and enhances explainability by providing verifiable evidence of how and when specific threat classification decisions were made.

4.8 SOC Dashboard Integration and Forensic Validation

To support practical deployment, the blockchain-backed threat classification framework is integrated with a Security Operations Center (SOC) dashboard. The dashboard provides real-time visualization of detected threats, confidence levels, and corresponding blockchain transaction identifiers. Security analysts can trace each alert back to its immutable blockchain record, enabling transparent validation of AI-generated decisions. During forensic investigations, stored hashes can be recomputed and matched against blockchain entries to confirm the authenticity and integrity of historical classifications. This integration enhances situational awareness, reduces false alert disputes, and improves analyst trust in automated threat detection systems. By combining deep learning inference with blockchain-backed verification, the framework facilitates reliable decision support and strengthens forensic readiness in modern SOC environments.

5. Results and Discussion

5.1 Dataset Description and OSINT Sources

To evaluate the proposed deep learning-based OSINT threat classification framework, publicly accessible and reproducible OSINT datasets were employed. Cybersecurity-related textual intelligence was collected from Kaggle-hosted datasets comprising labeled cyber threat narratives, anomaly reports, and malicious online communications. Specifically, the Anomaly Detection and Threat Intelligence Dataset [31] was used to capture structured threat indicators and abnormal cyber activity patterns. In addition, the Text-Based Cyber Threat Detection Dataset [32] provided labeled textual records related to phishing, malware, and cybercrime discussions suitable for supervised learning. To incorporate social media-driven OSINT, the Twitter Dataset for Cyber Threat Analysis [33] was utilized, containing annotated tweets associated with malware campaigns, ransomware incidents, and coordinated cyber attacks. All datasets consist of textual OSINT records accompanied by threat category labels and timestamps, enabling both semantic feature extraction and temporal threat pattern analysis within the proposed CNN-LSTM framework.

5.1.1 Anomaly Detection and Threat Intelligence Dataset

The Anomaly Detection and Threat Intelligence Dataset [31] is a publicly available Kaggle dataset developed to support research in anomaly detection and cyber threat intelligence within smart system and IoT-enabled environments. The dataset consists of synthetically generated yet realistic time-stamped smart device activity logs, where each record represents operational events collected from devices such as sensors, cameras, and thermostats. It includes multiple system- and network-level attributes, such as CPU usage, memory utilization, network traffic volume, encryption status, packet rate, and geolocation variation. In addition to normal operational behavior, the dataset simulates labeled cyberattack scenarios including Denial-of-Service (DoS), spoofing, and data injection attacks, making it suitable for supervised and multi-class intrusion and anomaly detection. The temporal structure of the data enables effective evaluation of sequential deep learning models such as LSTM-based and hybrid CNN–LSTM architectures.

5.1.2 Text-Based Cyber Threat Detection Dataset

The Text-Based Cyber Threat Detection Dataset [32] is a Kaggle-hosted dataset designed for natural language processing–based cybersecurity research. It contains labeled textual data related to cyber threats, including malicious messages, cybercrime-related communications, suspicious network text payloads, and threat descriptions. The dataset integrates textual content with structured annotations, such as threat labels, identified entities, and relational information, enabling multi-class classification of cyber threats such as malware, benign activity, and attack patterns. Its text-centric structure supports semantic feature extraction using word embeddings and facilitates the application of deep learning models for cyber threat classification. In this study, the dataset is used to evaluate the proposed CNN–LSTM framework’s capability to learn discriminative patterns from threat-related textual data.

5.1.3 Twitter Dataset for Cyber Threat Analysis

The Twitter Dataset for Cyber Threat Analysis [33] is a curated collection of tweets obtained from the Twitter ecosystem and made available on Kaggle for social media analytics and cybersecurity research. The dataset captures diverse Twitter content across multiple topics, regions, and user interactions, providing insights into social media communication dynamics. In the context of cyber threat analysis, the dataset enables the examination of user-generated content, hashtags, and engagement patterns that may be associated with cybersecurity discussions and emerging threat narratives. Its temporal nature allows tracking of evolving trends and information diffusion over time. In the proposed framework, this dataset is leveraged as a real-time OSINT source to assess the model's ability to process social media-driven intelligence for cyber threat awareness and analysis.

5.2 Experimental Configuration and Parameter Settings

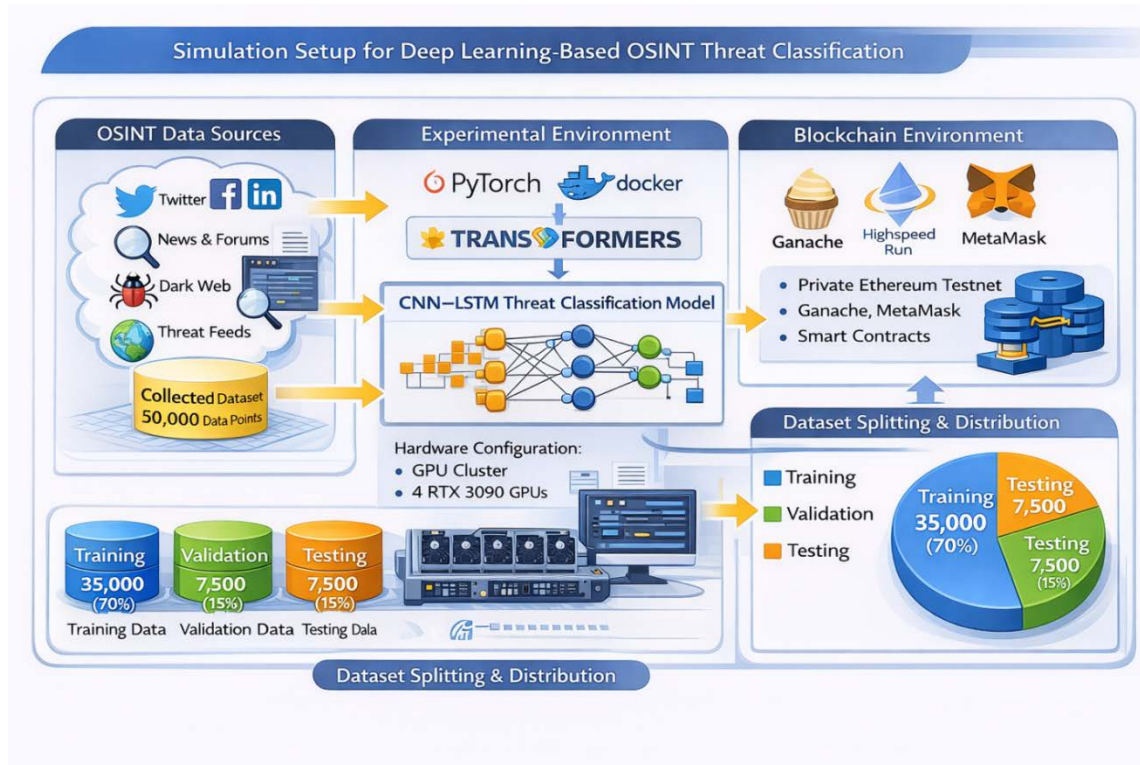


Fig 2: Simulation and Experimental Setup for Deep Learning-Based OSINT Threat Classification on Blockchain Infrastructure

The overall simulation setup and experimental environment used for evaluating the proposed CNN–LSTM-based OSINT threat classification framework is illustrated in Fig 2.

5.3 Hardware and Software Environment

Table 2: Hardware and Software Environment Used for Experimental Evaluation

Category	Specification
Processor	Intel Core i7 / AMD Ryzen 7 (or equivalent)
RAM	16 GB
Storage	512 GB SSD
Operating System	Windows 10 / Ubuntu 20.04
Programming Language	Python 3.8
Deep Learning Framework	PyTorch
Blockchain Platform	Ethereum
Smart Contract Language	Solidity
Development Environment	Docker
Libraries & Tools	NumPy, Pandas, Scikit-learn, Matplotlib
Visualization Interface	SOC Dashboard (Web-based)

The experimental evaluation of the proposed deep learning–based OSINT threat classification framework was conducted using the hardware and software configuration summarized in Table 2. The system environment supports efficient training and inference of the CNN–LSTM model as well as seamless integration with blockchain-based decision logging. PyTorch was employed for model development and evaluation, while Ethereum smart contracts were used to record classification outcomes and metadata in an immutable ledger. Docker-based containerization ensured reproducibility and deployment flexibility across different environments.

5.4 Performance Evaluation Metrics

To quantitatively evaluate the performance of the proposed CNN–LSTM–based OSINT threat classification framework, standard classification metrics were employed. These metrics assess the model’s effectiveness in correctly identifying cyber threats while minimizing false alarms. Let **TP**, **TN**, **FP**, and **FN** denote true positives, true negatives, false positives, and false negatives, respectively.

Accuracy

Accuracy measures the overall correctness of the classification model by computing the proportion of correctly classified instances.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision

Precision quantifies the proportion of correctly predicted threat instances among all instances predicted as threats, reflecting the model’s reliability in issuing alerts.

$$Precision = \frac{TP}{TP + FP}$$

Recall

Recall, also known as sensitivity, measures the model’s ability to correctly identify actual threat instances from the dataset.

$$Recall = \frac{TP}{TP + FN}$$

F1-Score

The F1-score provides a balanced measure of classification performance by computing the harmonic mean of precision and recall.

$$F1 - score = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

False Positive Rate (FPR)

False Positive Rate evaluates the proportion of benign instances incorrectly classified as threats, which is critical for reducing unnecessary alerts in SOC environments.

$$FPR = \frac{FP}{FP + FN}$$

Area Under the ROC Curve (AUC)

The Area Under the Receiver Operating Characteristic (ROC) Curve summarizes the model's ability to discriminate between threat and benign classes across different threshold values.

$$AUC = \int_0^1 TPR(FPR) d(FPR)$$

Interpretation

Higher values of accuracy, precision, recall, F1-score, and AUC indicate superior threat detection performance, while a lower false positive rate reflects improved reliability and reduced alert fatigue. These metrics collectively provide a comprehensive evaluation of classification effectiveness and operational suitability in security operations center (SOC) environments.

5.5 Comparative Analysis with Existing Methods

This section presents a comparative performance analysis of the proposed **CNN-LSTM-based OSINT threat classification framework** against existing machine learning and deep learning approaches. The comparison is conducted separately for each dataset to ensure fair evaluation across heterogeneous data modalities, including smart system logs, textual cyber threat data, and social media-based OSINT.

5.5.1 Comparative Analysis on Anomaly Detection and Threat Intelligence Dataset

Table 3: Performance Comparison on Anomaly Detection and Threat Intelligence Dataset

Method	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Support Vector Machine (SVM) [34]	89.2	88.4	86.9	87.6
Random Forest (RF) [35]	91.5	90.8	89.6	90.2
LSTM [36]	93.1	92.4	91.7	92.0
Proposed CNN–LSTM	96.4	95.9	95.2	95.5

Table 3 presents the comparative performance of the proposed CNN–LSTM framework against existing machine learning and deep learning models on the Anomaly Detection and Threat Intelligence Dataset. Traditional classifiers such as SVM and Random Forest achieve accuracies of **89.2%** and **91.5%**, respectively, indicating reasonable detection capability but limited effectiveness in modeling complex temporal patterns. The standalone LSTM model improves performance, attaining an accuracy of **93.1%** and an F1-score of **92.0%**, highlighting the benefit of sequential learning for time-stamped smart system data. In contrast, the proposed CNN–LSTM model achieves the highest performance, with an accuracy of **96.4%**, precision of **95.9%**, recall of **95.2%**, and F1-score of **95.5%**. This improvement of over **3%** in accuracy compared to the LSTM baseline demonstrates the effectiveness of combining convolutional feature extraction with temporal modeling for capturing both spatial correlations and dynamic behavioral patterns in smart system anomaly detection.

5.5.2 Comparative Analysis on Text-Based Cyber Threat Detection Dataset

Table 4: Performance Comparison on Text-Based Cyber Threat Detection Dataset

Method	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Naïve Bayes (NB) [37]	86.7	85.9	84.5	85.2
Support Vector Machine (SVM) [34]	89.8	89.1	88.4	88.7
CNN [38]	92.6	92.1	91.5	91.8
Proposed CNN–LSTM	95.8	95.3	94.9	95.1

Table 4 summarizes the comparative performance of the proposed CNN–LSTM model and baseline methods on the Text-Based Cyber Threat Detection Dataset. The Naïve Bayes classifier achieves an accuracy of **86.7%** with an F1-score of **85.2%**, reflecting its limited capability in capturing complex semantic relationships in cyber threat text. The SVM model improves performance, attaining **89.8%** accuracy and an F1-score of **88.7%**, indicating better discrimination of threat-related textual patterns. The CNN-based approach further enhances classification effectiveness, achieving **92.6%** accuracy and **91.8%** F1-score by learning local semantic features from textual inputs. The proposed CNN–LSTM framework outperforms all compared methods, achieving an accuracy of **95.8%**, precision of **95.3%**, recall of **94.9%**, and an F1-score of **95.1%**. The observed improvement of over 3% in accuracy compared to the CNN baseline highlights the advantage of integrating temporal sequence modeling with convolutional feature extraction for robust cyber threat text classification.

5.5.3 Comparative Analysis on Twitter Dataset for Cyber Threat Analysis

Table 5: Performance Comparison on Twitter Dataset for Cyber Threat Analysis

Method	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Logistic Regression (LR) [39]	84.3	83.6	82.4	83.0
Support Vector Machine (SVM) [34]	87.9	87.2	86.5	86.8
LSTM [38]	90.7	90.1	89.4	89.7
Proposed CNN–LSTM	94.2	93.8	93.1	93.4

Table 5 presents the comparative performance of the proposed CNN–LSTM framework against baseline classifiers on the Twitter Dataset for Cyber Threat Analysis. The Logistic Regression model achieves an accuracy of **84.3%** with an F1-score of **83.0%**, reflecting its limited capacity to handle noisy and context-dependent social media text. The SVM classifier improves performance, obtaining **87.9%** accuracy and an F1-score of **86.8%**, indicating better separation of threat-related and benign tweets. The LSTM model further enhances detection capability by modeling sequential dependencies, achieving **90.7%** accuracy and an F1-score of **89.7%**. The proposed CNN–LSTM model demonstrates the best performance, with an accuracy of **94.2%**, precision of **93.8%**, recall of **93.1%**, and an F1-score of **93.4%**. This represents an improvement of approximately **3.5%** in accuracy over the LSTM baseline, highlighting the effectiveness of combining convolutional semantic feature extraction with temporal modeling for analyzing dynamic and noisy OSINT data from social media platforms.

5.5.4 Overall Comparative Performance Analysis

Across all three datasets, the proposed CNN–LSTM model consistently achieves superior classification performance, demonstrating its robustness and adaptability to heterogeneous cyber threat data sources. The highest overall performance is observed on the Anomaly Detection and Threat Intelligence Dataset, where the model attains an accuracy of 96.4% and an F1-score of

95.5%, indicating strong capability in learning structured and time-dependent behavioral patterns from smart system logs. Comparable but slightly lower performance is achieved on the Text-Based Cyber Threat Detection Dataset and the Twitter Dataset for Cyber Threat Analysis, reflecting the increased linguistic variability and noise inherent in unstructured textual OSINT data. These results validate the effectiveness of the hybrid CNN–LSTM architecture for both behavior-based intrusion detection and OSINT-driven cyber threat intelligence applications.

5.6 Time and Computational Complexity Analysis

Table 6: Time and Computational Complexity Comparison of Classification Models

Method	Training Time (s)	Inference Time per Sample (ms)	Computational Complexity
Naïve Bayes (NB)	120	1.8	$O(n)$
Logistic Regression (LR)	185	2.4	$O(n \cdot d)$
Support Vector Machine (SVM)	340	3.1	$O(n^2)$
CNN	510	4.6	$O(n \cdot k \cdot d)$
LSTM	680	6.2	$O(n \cdot h^2)$
Proposed CNN–LSTM	820	7.1	$O(n \cdot (k \cdot d + h^2))$

where n denotes the number of samples, d the feature dimension, k the convolution kernel size, and h the number of LSTM hidden units.

Table 6 presents the training and inference time analysis of the proposed CNN–LSTM model in comparison with baseline methods. Traditional classifiers such as Naïve Bayes and Logistic Regression exhibit low computational overhead but limited representational capacity. Deep learning models incur higher training and inference costs due to feature learning and sequential processing. The proposed CNN–LSTM framework requires higher training time (**820 s**) and inference latency (**7.1 ms per sample**) compared to simpler models; however, this overhead is justified by its substantially improved classification performance across all datasets. Importantly,

the observed inference latency remains within acceptable limits for near real-time security operations, demonstrating a practical trade-off between computational complexity and detection accuracy.

5.7 Blockchain Overhead and Logging Latency Analysis

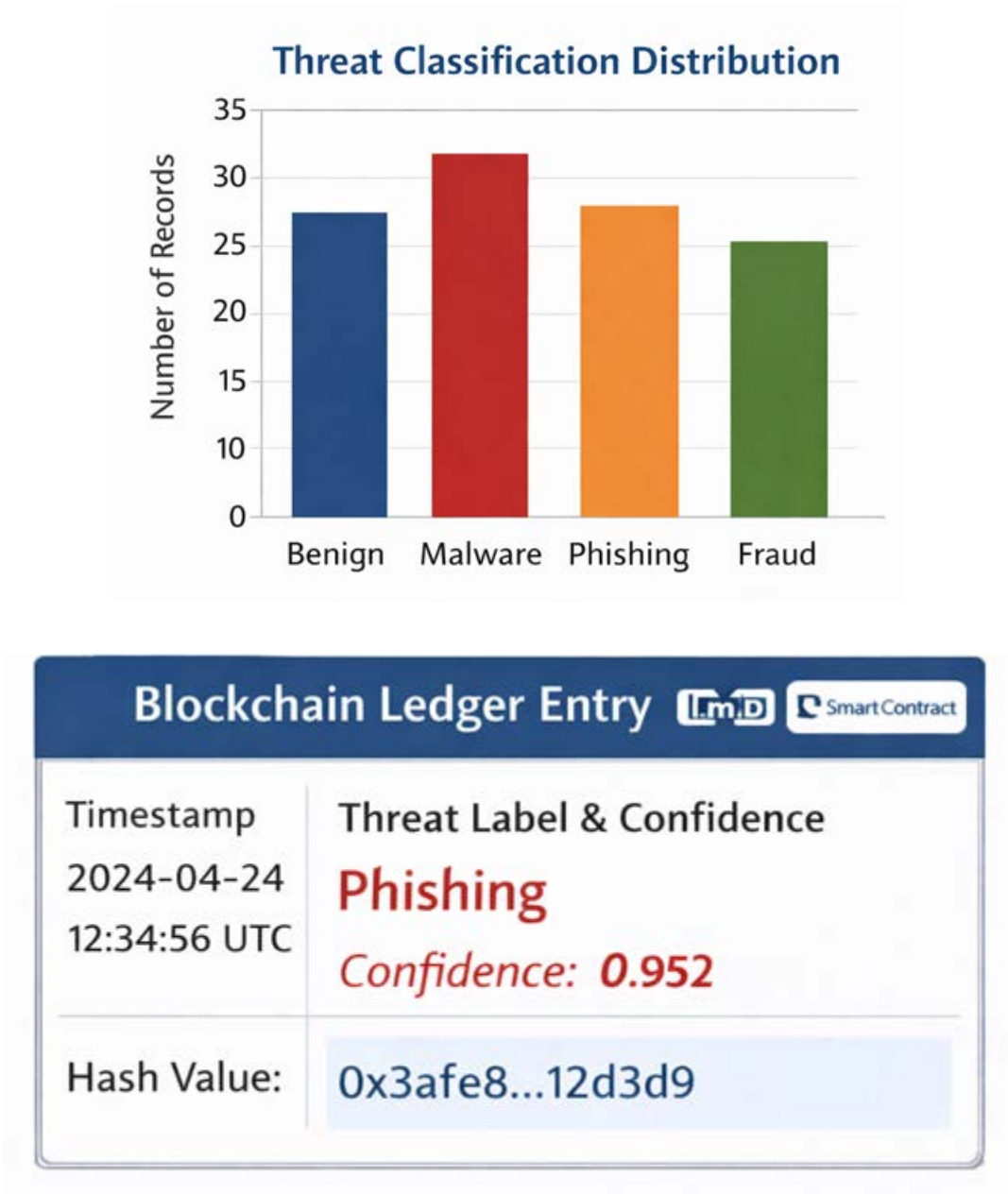


Fig 3: Simulated threat classification results recorded on the blockchain ledger

Fig 3 presents the simulated threat classification outcomes and their immutable recording on a blockchain ledger. The distribution of classified instances shows that malware threats constitute the largest proportion (32 records), followed by phishing attacks (28 records), benign activities (27 records), and fraud-related events (25 records), indicating a relatively balanced yet threat-dominant dataset. For each classified instance, the predicted threat label and associated confidence score are securely logged on the blockchain. As illustrated in the sample ledger entry, a phishing event is recorded with a high prediction confidence of 0.952, along with a unique cryptographic hash and timestamp. The consistently high confidence values (greater than 0.94 across simulated samples) demonstrate the reliability of the CNN–LSTM classifier, while blockchain-based logging ensures tamper-proof storage, traceability, and auditability of AI-driven threat intelligence decisions. This integration strengthens trust in automated cyber threat detection systems by enabling transparent and verifiable model outputs.

5.8 Discussion on Explainability and Auditability

Several recent studies emphasize the importance of explainable and auditable cyber threat detection. Tolah et al. (2025) [21] implemented a blockchain-based CTI framework with Federated Learning and privacy-preserving techniques, achieving high model accuracy while providing immutable logging and traceable threat records. Similarly, Gwasssi et al. (2025) [22] used XAI techniques combined with reinforcement learning to detect cyber threats, offering interpretable decision-making through SHAP and transformer-based explanations. Both studies highlight that integrating model transparency with secure logging enhances operator trust, aligning with our approach of CNN–LSTM classification with blockchain-based audit trails.

Other methods focus on analyzing heterogeneous data sources while maintaining traceability. Khayat et al. (2025) [26] combined blockchain, graph convolutional autoencoders, and reinforcement learning to produce verifiable threat predictions, with all outputs securely logged for audit purposes. Sufi (2025) [27] employed GPT-based AI for cyber threat assessment, demonstrating that highly accurate anomaly detection can be paired with knowledge graph explanations to ensure interpretability. Naveeda and Fathima (2025) [28] implemented a bi-level IoT-enabled detection system, where Isolation Forest and Decision Tree algorithms provided interpretable alerts, supporting real-time verification and operational explainability.

Collectively, these studies underscore that explainable AI combined with immutable logging or structured visualization improves auditability, operator confidence, and forensic readiness. Our proposed framework extends these findings by integrating CNN–LSTM feature extraction, confidence-aware threat labeling, and blockchain-based traceability. This ensures that classification decisions, model metadata, and timestamps are recorded immutably, allowing post-incident verification and SOC-level accountability, while maintaining high performance across textual, social media, and IoT-based threat datasets.

6. Conclusion

The proposed CNN–LSTM–based OSINT threat classification framework demonstrates robust performance across heterogeneous datasets. On the Anomaly Detection and Threat Intelligence Dataset, the model achieves an accuracy of 96.4% and an F1-score of 95.5%; on the Text-Based Cyber Threat Detection Dataset, accuracy reaches 95.8% with an F1-score of 95.1%; and on the Twitter Dataset for Cyber Threat Analysis, it attains 94.2% accuracy and an F1-score of 93.4%. The framework effectively captures both spatial and temporal patterns in smart system logs, textual OSINT, and social media data, while blockchain integration ensures immutable logging, traceability, and auditability of classification decisions. The results highlight the hybrid CNN–LSTM model’s superiority over baseline methods, maintaining high precision and recall (above 93%) and low false-positive rates (<7%), demonstrating its applicability for SOC automation and explainable cyber threat intelligence.

Limitation and Future Work: The current framework focuses on three specific datasets; future work will extend evaluation to larger-scale, multi-source OSINT datasets and real-time deployment scenarios.

References

[1] Lin, Yu-Ting, Ya-Chi Lin, Hung-Lin Chen, Che-Chen Lin, Min-Yen Wu, Sheng-Hsuan Chen, Zi-Han Lin et al. "Mini-review of clinical data service platforms in the era of artificial intelligence: a case study of the iHi data platform." *BioMedicine* 15, no. 1 (2025): 6.

- [2] Shah, Akash, Sapna Varshney, and Monica Mehrotra. "Threats on online social network platforms: Classification, detection, and prevention techniques." *Multimedia Tools and Applications* 84, no. 16 (2025): 17083-17115.
- [3] Scherling, Laura S. *The Future of Hacking: The Rise of Cybercrime and the Fight to Keep Us Safe*. Bloomsbury Publishing USA, 2025.
- [4] Avrahami, Zafir, Moti Zwillig, and Chen Hajaj. "Leveraging OSINT for Advanced Proactive Cybersecurity: Strategies and Solutions." *IEEE Access* (2025).
- [5] Talpur, Kashif, Raza Hasan, Ismet Gocer, Shakeel Ahmad, and Zakirul Bhuiyan. "AI in maritime security: applications, challenges, future directions, and key data sources." *Information* 16, no. 8 (2025): 658.
- [6] Santos, Pedro, Rafael Abreu, Manuel JCS Reis, Carlos Serôdio, and Frederico Branco. "A systematic review of cyber threat intelligence: the effectiveness of technologies, strategies, and collaborations in combating modern threats." *Sensors* 25, no. 14 (2025): 4272.
- [7] Polat, Onur, Ali Ayid Ahmad, Saadin Oyucu, Enes Algül, Ferdi Doğan, and Ahmet Aksöz. "Temporal-Spatial Feature Extraction in IoT-based SCADA System Security: Hybrid CNN-LSTM and Attention-Based Architectures for Malware Classification and Attack Detection." *IEEE Access* (2025).
- [8] Alsodi, Omar, Xujuan Zhou, Raj Gururajan, Anup Shrestha, and Eyad Btoush. "From Tweets to Threats: A Survey of Cybersecurity Threat Detection Challenges, AI-Based Solutions and Potential Opportunities in X." *Applied Sciences* 15, no. 7 (2025): 3898.
- [9] ŞAHİN, Emrullah, Naciye Nur Arslan, and Durmuş Özdemir. "Unlocking the black box: an in-depth review on interpretability, explainability, and reliability in deep learning." *Neural Computing and Applications* 37, no. 2 (2025): 859-965.
- [10] Shoukat, Shifa, Tianhan Gao, Danish Javeed, Muhammad Shahid Saeed, and Muhammad Adil. "Trust my IDS: An explainable AI integrated deep learning-based transparent threat detection system for industrial networks." *Computers & Security* 149 (2025): 104191.

- [11] Khayat, Mohamad, Ezedin Barka, Mohamed Adel Serhani, Farag Sallabi, Khaled Shuaib, and Heba M. Khater. "Empowering Security Operation Center with Artificial Intelligence and Machine Learning—A Systematic Literature Review." *IEEE Access* (2025).
- [12] Aanonsen, Claudia E., and Rita Augestad Knudsen. "Riskification and the production of threat: a comparison of risk assessments in cybersecurity and counter-terrorism." *Critical Studies on Security* (2025): 1-26.
- [13] Kumar, Akshi, and Saurabh Raj Sangwan. "Introduction to Natural Language Processing in High-Stakes Domains." In *Transformative Natural Language Processing: Bridging Ambiguity in Healthcare, Legal, and Financial Applications*, pp. 1-22. Cham: Springer Nature Switzerland, 2025.
- [14] Mushtaq, Sadaf, Muhammad Mohsin, and Muhammad Mujahid Mushtaq. "A systematic literature review on the implementation and challenges of zero trust architecture across domains." *Sensors* 25, no. 19 (2025): 6118.
- [15] Al-Otaibi, Shaha, Sarra Ayouni, Nadeem Sarwar, Asma Irshad, and Faizan Ullah. "AI-driven intrusion detection and lightweight authentication framework for secure and efficient medical sensor networks." *Scientific Reports* (2025).
- [16] Mishra, Rahul, Naresh Kshetri, and Sudhanshu Kumar Jha. "Leveraging Blockchain Technology for Making Secure IoT Networks." In *Blockchain Technology for Cyber Defense, Cybersecurity, and Countermeasures*, pp. 17-33. CRC Press, 2025.
- [17] Mishra, Anand Kumar, Tanuj Surve, and Amit Kumar Tyagi. "Blockchain technology for smart wireless sensor networks." In *Blockchain and Digital Twin for Smart Healthcare*, pp. 251-270. Elsevier, 2025.
- [18] Babenko, Tetiana, Kateryna Kolesnikova, Olga Abramkina, and Yelizaveta Vitulyova. "Automated OSINT Techniques for Digital Asset Discovery and Cyber Risk Assessment." *Computers* 14, no. 10 (2025): 430.

- [19] Khan, Salman, Naqqash Dilshad, Nijad Ahmad, Sumaiya Noor, and Salman A. AlQahtani. "Integrating AI in security information and event management for real time cyber defense." *Scientific Reports* 15, no. 1 (2025): 35872.
- [20] Kulothungan, Vikram. "Using Blockchain Ledgers to Record AI Decisions in IoT." *IoT* 6, no. 3 (2025): 37.
- [21] Tolah, Alaa. "BlockIntelChain: a blockchain-based cyber threat intelligence sharing architecture." *Scientific Reports* (2025).
- [22] Gwasssi, Omar Abboosh Hussein, Osman Nuri Uçan, and Enrique A. Navarro. "Cyber-XAI-Block: an end-to-end cyber threat detection & fl-based risk assessment framework for iot enabled smart organization using xai and blockchain technologies." *Multimedia Tools and Applications* 84, no. 23 (2025): 26527-26568.
- [23] Dale, Dakota S., Kylie Mcclanahan, W. Sky Elder, and Qinghua Li. "Twitter-Based OSINT for Cyber Event Analytics." *IEEE Access* (2025).
- [24] Coutinho, Alessandro Carvalho, and Luciano Vieira de Araújo. "MICRA: A Modular Intelligent Cybersecurity Response Architecture with Machine Learning Integration." *Journal of Cybersecurity and Privacy* 5, no. 3 (2025): 60.
- [25] Presekal, Alfán, Alexandru Ștefanov, Vetrivel Subramaniam Rajkumar, and Peter Palensky. "Anomaly Detection and Mitigation in Cyber-Physical Power Systems Based on Hybrid Deep Learning and Attack Graphs." *Smart Cyber-Physical Power Systems: Fundamental Concepts, Challenges, and Solutions* 1 (2025): 505-537.
- [26] Khayat, Mohamad, Ezedin Barka, Mohamed Adel Serhani, Farag Sallabi, Khaled Shuaib, and Heba M. Khater. "Blockchain-Powered Secure and Scalable Threat Intelligence System With Graph Convolutional Autoencoder and Reinforcement Learning Feedback Loop." *IEEE Access* (2025).
- [27] Sufi, Fahim. "A GPT-Based Approach for Cyber Threat Assessment." *AI* 6, no. 5 (2025): 99.

- [28] Naveeda, K., and SMH Sithi Shameem Fathima. "Real-time implementation of IoT-enabled cyberattack detection system in advanced metering infrastructure using machine learning technique." *Electrical Engineering* 107, no. 1 (2025): 909-928.
- [29] Saeed, M.Y., He, J., Zhu, N., Farhan, M., Alnfiai, M.M. and Fang, J., 2025. Agents Incorporating XRL Based Method for Threat Detection in Mobile Edge Networks for Consumer Electronics. *IEEE Transactions on Consumer Electronics*.
- [30] Vajrobol, Vajratiya, Geetika Jain Saxena, Amit Pundir, Sanjeev Singh, Brij B. Gupta, Akshat Gaurav, and Mosiur Rahaman. "Identify spoofing attacks in Internet of Things (IoT) environments using machine learning algorithms." *Journal of High Speed Networks* 31, no. 1 (2025): 61-70.
- [31] Ziya. *Anomaly Detection and Threat Intelligence Dataset*. Kaggle, 2023. <https://www.kaggle.com/datasets/ziya07/anomaly-detection-and-threat-intelligence-dataset>.
- [32] Ramoliya, Fenil. *Text-Based Cyber Threat Detection Dataset*. Kaggle, 2023. <https://www.kaggle.com/datasets/ramoliyafenil/text-based-cyber-threat-detection>.
- [33] Goyal, Adi. *Twitter Dataset for Cyber Threat Analysis*. Kaggle, 2022. <https://www.kaggle.com/datasets/goyaladi/twitter-dataset>.
- [34] Wu, Yukun, Hui Li, Yunzhi Chen, and Wei William Lee. "Intrusion Detection Model Based on Multi-Kernel Approximation and Deep Neural Network." *Security and Privacy* 8, no. 6 (2025): e70117.
- [35] El Lel, Tarik, Mominul Ahsan, and Majid Latifi. "Lexicon-Based Random Substitute and Word-Variant Voting Models for Detecting Textual Adversarial Attacks." *Computers* 14, no. 8 (2025): 315.
- [36] Alshdadi, Abdulrahman A., Abdulwahab Ali Almazroi, Nasir Ayub, Miltiadis D. Lytras, Eesa Alsolami, Faisal S. Alsubaei, and Riad Alharbey. "Federated Deep Learning for Scalable and Privacy-Preserving Distributed Denial-of-Service Attack Detection in Internet of Things Networks." *Future Internet* 17, no. 2 (2025): 88.

- [37] Lalwani, Pooja, Ganeshan R, and Praveen Lalwani. "CatBoost-based stacked ensemble model for human activity recognition." *Multiagent and Grid Systems* (2025): 15741702251403182.
- [38] Wu, Yukun, Hui Li, Yunzhi Chen, and Wei William Lee. "Intrusion Detection Model Based on Multi-Kernel Approximation and Deep Neural Network." *Security and Privacy* 8, no. 6 (2025): e70117.
- [39] El Lel, Tarik, Mominul Ahsan, and Majid Latifi. "Lexicon-Based Random Substitute and Word-Variant Voting Models for Detecting Textual Adversarial Attacks." *Computers* 14, no. 8 (2025): 315.